

A Practical Framework for Ransomware Detection and Mitigation Strategies

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Abstract—Ransomware attacks have emerged as one of the most significant cyber threats, particularly targeting financial organizations, with incidents rising from 48% in 2020 to 64% in 2023. Despite the increasing frequency of ransomware attacks, existing detection mechanisms struggle to effectively identify and mitigate new and evolving ransomware variants, leaving financial organizations highly vulnerable to cyber extortion. This paper presents a comprehensive detection algorithm aimed at well-known ransomware families such as WannaCry, NotPetya, Locky, and Ryuk, alongside a practical strategy for detection and prevention. Additionally, the proposed ransomware detection system examines suspicious strings, heuristic rules, file structures, executable file extensions, and Command and Control (CnC) server patterns. Evaluated against a dataset of 144 files, including both ransomware samples and benign documents, the system achieves an accuracy of over 81%, a precision of 82.86%, and a recall of 80.56%. Finally, the study outlines a multi-layered mitigation strategy encompassing prevention, detection, incident response, and recovery procedures.

Keywords—ransomware detection, cybersecurity, mitigation strategies

I. INTRODUCTION

Ransomware damages global business and encrypts important data until a ransom is paid. 64% of financial organizations are being attacked by ransomware with an estimated \$2.23 million to fully recover after ransomware attack excluding ransom payments in only 2023 year [1]. In the same year, only 14% of the finance companies were able to prohibit the attack before the data was encrypted by the attacker and in 25% of attacks that data was stolen by them. Overall, Sophos's 2023 report was based on 3,000 cybersecurity and IT professionals working for businesses in 14 countries. Results for the financial services sector were derived from a subset of 336 respondents who were especially involved in that industry. Other financial services, higher education have 79 %, construction 71 % federal government, media and entertainment 70 %, and manufacturing which list out with 56 % hit by ransomware attacks.

Large strains of ransomware have caused damage to many kinds of industries including finance and healthcare. This research will provide a comprehensive analysis of the most significant ransomware families, for example, WannaCry, NotPetya, Locky, and Ryuk. The review will gain crucial insights into how ransomware penetrates defenses through methodical analysis of the infection strategies, encryption methods, and evasion techniques them.

This research will focus on analyzing the behavior patterns of ransomware because behavior patterns exhibited by different ransomware families during the infection, encryption, and ransom process. For the hypothesis it, some ransomware variants may show their unique behavioral characteristics such as different encryption techniques or payment methods which might affect their impact and detection capabilities. Additionally, the research also provides a comprehensive mitigation plan from prevention to post-event of ransomware attacks to reduce the damage caused by ransomware attacks. Therefore, an ideal system will be proposed to assist in preventing the successful downloading of ransomware. The mitigation plan is important for preventing post-event ransomware attacks. By analyzing the ransomware, it can enhance the awareness of ransomware attacks and better defend against these cyberattacks.

II. LITERATURE REVIEW

A. Overview of Ransomware

Ransomware is a type of malicious software that encrypts victims' or institutional targets' files or locks them out of their computer system or network until the attacker is paid a ransom [2, 3]. The main threat of ransomware is to focus on functions by encrypting the target's files and requesting a ransom in exchange for a decryption key. The attackers usually use phishing emails, malicious attachments, or software vulnerabilities to infiltrate the target's computer system. After the ransomware is installed, the virus encrypts the target's data, making it useless and unreadable, and displays a ransom note with instructions on how to decrypt it by paying the attacker with cryptocurrency. If the ransom is paid, the attacker will provide the decryption key to allow the target to regain accessibility to their files and folders. However, there is no guarantee that the data will be returned, and in some cases, the attacker may not provide the decryption key although you pay the attacker. [4]

B. Four Critical Component of Ransomware

After that, the entire ransomware assault method is made up of numerous components, so we focus on 4 components which include Encryption algorithm, Recovery prevention techniques, C2 communication and target audience. [3]

First is the encryption algorithm. The latest ransomware uses hybrid cryptosystems where malware generates host-generated symmetric and sub-symmetric keys, such as RSA and AES, and uses the AES keys to encrypt data. The AES keys are encrypted using the sub-RSA key, and the sub-RSA key is encrypted using the built-in key. In short, we can

observe that the encryption algorithm became more complex from the start of the ransomware attack until now.

Second, even when data is encrypted using any of the previous methods, some of it can still be retrieved via third-party applications or system volume shadow copies. The newer ransomware strains proceed to erase the volume shadow copies and the remaining files. Typically, `vssadmin.exe` is used to remove volume shadow copies, and the files are either removed manually by deleting the files' meta-data information and directory structures, or they are removed by another method. Therefore, the newer ransomware strains to erase volume shadow copies and inhibit data recovery by overwriting the original target files after encryption. [5, 6]

The third of the components of the ransomware is C2 communication. Ransom payments are handled by C2, typically through the Bitcoin network. C2 communication is often referred to as Command-and-Control communication and it is a critical component of ransomware operations and for other cyberattacks. It acts as a connection between the attacker's infrastructure and the infected or compromised devices. Although the attacker may own the C2 infrastructure, it is typically a botnet of infected devices hiding behind an anonymous network. Some newer ransomware strains are the real payload absent, and the ransomware is spread via scripts. Rather, the script makes communication with the C2 server to obtain the payload and related keys as soon as it becomes infected. By doing this, it becomes more challenging to identify the ransomware early on while just the script is executing. Furthermore, some ransomware payloads like Cryptowall, are unable to initiate the process of contacting the C2 server for additional instructions. As a result, C2 communications are now a crucial feature of the most recent ransomware strains. In short conclusion, the benefits of the new strains of ransomware include increasing the ability of stealth and evasion by not including the actual ransomware payload, reducing footprint by keeping the payload off the infected system, and lastly increasing the control and extortion like Cryptowall designed to be highly controlled by the attackers. [7]

The last component of the ransomware is the target audience. Ransomware has changed from randomly attacking internet users to now focusing on businesses because these targets are more profitable for them. According to recent trends, attackers are increasingly using exploit kits and exploiting vulnerabilities in internet-facing remote access services like Remote Desktop Protocol (RDP) to target enterprise systems. Most recent ransomware strains such as WannaCry, SamSam, Erebus, and others have worm-like qualities that allow them to travel across an information system's whole network structure and infect any host that is found to be vulnerable. Therefore, this leads to attacks on online backup systems.

C. Common Tactics and Techniques Employed by Ransomware Families

A comparative analysis of the tactics and techniques employed by four ransomware families: WannaCry [8], Ryuk [9], NotPetya [10], and Locky [11] has been conducted. These ransomware variants demonstrate distinct methods for initial access, privilege escalation, network infection, defense evasion, persistence, file encryption, and system recovery inhibition. WannaCry and NotPetya primarily exploit the

EternalBlue vulnerability for network propagation, while Locky leverages phishing emails and exploit kits. Ryuk employs a more targeted approach through a "triple threat" campaign involving delivery, encryption, and extortion. All four employ advanced techniques to evade detection and maintain persistence, such as modifying registry entries or creating scheduled tasks. Additionally, their encryption strategies vary, with WannaCry, Ryuk, and Locky using AES and RSA, while NotPetya destroys the Master File Table. These diverse methods highlight the evolving and sophisticated nature of ransomware attacks.

D. The Effectiveness of Pass Mitigation Strategies

The main components of the mitigation plan that have been put into practice are network segmentation, end-user training, backup and recovery, and continually updating your system software to the most recent version.[3]

The benefit of always updating the system to the latest version is improving your system's vulnerability. This is always highly effective for the organizations that keep their systems up to date. On the other hand, those who fail to patch expose themselves to significant risk. [12]

Moreover, end-user education can prevent users from falling for phishing scams. This is because malicious links or documents make up most phishing campaigns. Users can learn to recognize suspicious emails and attachments by increasing awareness about the common tactics used by ransomware attackers. [13]

After that, network segmentation effectively limits the infection area during the ransomware disaster. Implementing network segmentation can prevent the lateral movement of ransomware within an organization or a system. Therefore, it is effective for limiting the impact of ransomware within an organization once it has infiltrated the organization's network. [14]

Finally, regular backup is a crucial strategy for data recovery because it allows organizations to restore their systems without paying ransom. Thus, it is necessary to maintain up-to-date and offline backups because it enables recovery in a short period without paying the ransom. [15]

III. IMPLEMENTATION PROCESS

In this research, we will develop a comprehensive mitigation plan and evaluate the performance of the proposed system in terms of accuracy and response time. All ransomware samples used to evaluate the accuracy of the proposed system were obtained from GitHub repositories.

A. Overall Mitigation Plan

It is essential to build a comprehensive mitigation plan against ransomware to avoid and reduce many financial losses from ransomware attacks.

First, the prevention strategy is required to start from the successful ransomware downloaded in the computer, so detection and classification is the most important plan in the whole strategy. It is required to implement machine learning to detect and classify the ransomware to prevent it from being downloaded or executed based on behaviour, encryption methods, and the presence of the suspicious string in the ransomware folder. It includes network-based, host-based, and forensic characterization [16]. Second, the utilization of log services to analyze Command and Control traffic is

suggested. A log service is a system component responsible for collecting, storing, and analyzing the log messages that are generated by software and hardware components within a computer network. So, log services can detect ransomware Command and Control servers by analyzing network traffic, signature-based detection, and its behavior. It involves monitoring DNS requests, and web proxy logs to detect and block the known ransomware domains [17]. Therefore, it shows the important technique used in the detection of ransomware.

Besides, a proper incident response plan is also required to prevent more serious damage and attacks during the event of ransomware attacks. The plan should include immediate isolation of the affected system to prevent it from spreading, good communication within the organization or affected person, and legal and regulatory compliance on understanding any obligations regarding data breaches. This step is greatly effective in reducing the damage of attack damage and speeding up recovery to a healthy status [18].

Fourth, the backup of the data and folder plays an important role in the data recovery process after suffering a ransomware attack. The primary purpose is to ensure the recovery of data in the event of data loss or corruption. During the event of a ransomware attack, the files are encrypted, so having a recent backup can restore and get back to a healthy status without paying the ransom. Therefore, it is greatly efficient in mitigating the impact of ransomware attacks [18].

Furthermore, the prevention of further ransomware attacks is also important to prevent or reduce the probability of the attack happening again. The plan includes the steps to investigate ransomware attacks, for example, to use forensics and tools to reconstruct the stages of the ransomware attack life cycle [19].

In a nutshell, a comprehensive mitigation plan is greatly effective in cost reduction, and the organization's internet security. Therefore, the ransomware detector will play an important role as the front-line defender with the firewall, and security software while against the ransomware attacks.

B. Overall System Flow

Figure 1 illustrates the flow diagram of the detector. To begin, the user provides the correct file path for the file to be scanned by the detector and executes the command. The file then undergoes a verification process, checking both its executable file extension and file structure content. If the file is identified as containing an executable file extension, the detector displays a message notifying the user of this finding. Next, the file is processed through the heuristic rules checking function. If the file fails this check, the detector will issue an alert message to the user. Once the heuristic rules check is completed, the file undergoes a suspicious string analysis. If any suspicious strings are detected, the detector generates an alert to notify the user. The last function is CnC server checking which will search for CnC server pattern in the file. After completely going through those functions, the detector will calculate the advice score and generate the general advice according to the advice score to the user.

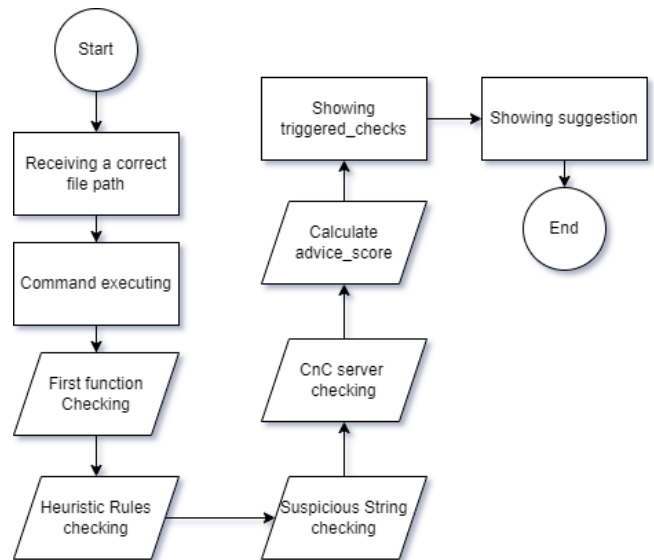


Fig. 1. Overall System Flow Diagram

C. Algorithm Detail

First, the detector will check the file extension, especially focusing on the executable file extension patterns based on and suspicious file structures pattern based on which are always used in ransomware executable files. These two functions work together to identify potential ransomware files or directories by testing their extensions and internal structure.

Next, for the checking heuristic function, it will check if the file contains code fragments associated with the encryption loop and encryption key. For the encryption key pattern which is used from this function to detect in the heuristic rule's function. Besides, the file modification function pattern will be detected by the presence of Windows programming for file operations and process creation.

For checking suspicious string function, it will check whether the file contains suspicious strings associated with viruses. The suspicious string will be found in the folder or zip file content of Word for known ransomware file extensions, generic ransom notes, encryption operations, and system manipulation like calling the Windows API function.

The window API function calls can be exploited by ransomware or during cyberattacks to perform actions example scheduling tasks, encrypting data, modifying registry values, and sending HTTP requests.

For checking the known ransomware file extensions, the presence of them is potentially hazardous to the user device and commonly appended to them during the encryption process. The detector can detect potential ransomware-infected files by looking at the filenames for the patterns. [13]

For checking the generic ransom notes pattern, some example patterns always appear with ransomware attacks. For example, the phrase 'Your data has been locked' likely appears after the ransomware successfully being exploited in the user's devices to inform the user that the data is locked. [3]

For the function of checking the CnC pattern, it is used to identify potential Command and Control server patterns within files, or the ZIP archive of the pattern.

The last function is designed to perform calling the function from above and giving user general advice based on the results of triggered checks and the advice score. For the

advice score, the detector provides the advice score to each function being detected.

IV. DATASETS SOURCES AND EXPERIMENT OUTCOMES

A. Datasets

In this experiment, the dataset is employed including of 144 files samples that come from GitHub repositories [20] and the official learning website. The 72 ransomware files like basic worm-like ransomware, RSA encrypted file locking ransomware, and those requiring command and control server activation, also have another 72 normal and safe documents like presentation deck and photos. The various ransomware families aim to reflect the complexity of malicious software and will provide a robust foundation for the experiment.

B. Experiment Result

The experiment result is based on the detector of advice_score. The advice_score is calculated based on the file being detected as the problem it has. The advice_score is shown below in Table I.

TABLE I. ADVISE SCORE TABLE

Metric	Advice Score
Suspicious file structure	1
Suspicious executable file extension	2
Heuristic rule not pass	2
Suspicious String found	2
CnC server pattern found	2

Table II shows how many of the files belong to not pass and pass. First, the file will be scanned through 4 functions. It is scanned starting from the check file extension. If it is detected that a suspicious file extension, it will go through to the checking file structure-function. If the file is detected only suspicious file structure, there will be an increment of advice_score of 2. If the file has a suspicious file structure, there will only be an increment of the advice_score of 1. After that, if the file detects that the heuristic rule does not pass, there will be an increment of advice_score of 2. Next, the suspicious strings contained in the file will also have an increment of 2 for the advice_score. Lastly, the file contains the CnC server pattern, it will have also an increment of 2 to the advice_score.

TABLE II. TABLE FOR THE RESULT

Metric	Quantity
Total folder being detected	144
Safe file	72
Ransomware file	72
Not pass (safe file)	14
Not pass (Ransomware)	58
Pass (Ransomware)	12
Pass (Safe File)	60

After the system finishes scanning the file, a total advice_score will be calculated. If the total advice score is 0, the file is clean, and it is safe to execute or open. If the file gets an advice score of 1, the user needs to open it with caution. If the file gets an advice score of 2 and 3, the user better understands the file before opening or executing it. If the file

gets a score is more than 3, the system suggests it deletes the malicious folder instead of opening or executing it.

The file is calculated as a safe file or pass when it gets a 0 or 1 advice score. When the advice score is more than 1, the file belongs to not safe or not pass.

As illustrated in Figure 2, the .lnk file is the most frequently identified file extension with ransomware characteristics. Both .dll and .lnk file extensions exhibit a 100% rate of being incorrectly flagged as unsafe, even when they are safe. This data will be utilized to calculate the false positive rate.

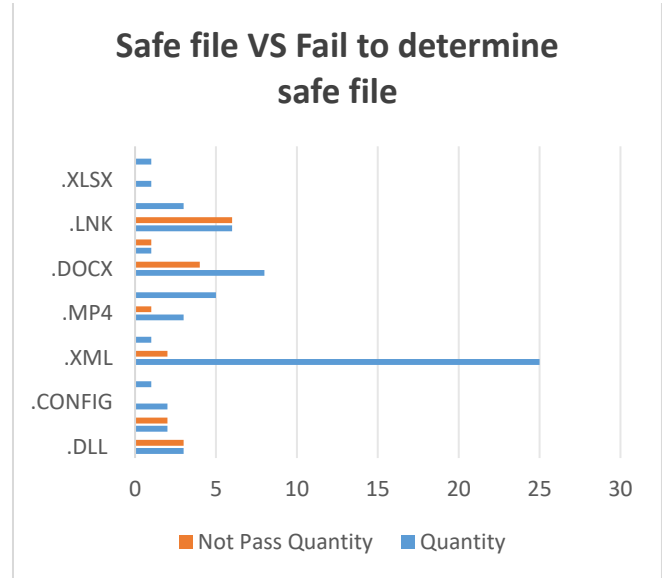


Fig. 2. Safe file VS Fail to determine safe file

C. Confusion Matrix

Table III and Eq. (1) – Eq. (4) show the performance matrices that are used to calculate the evaluation of performance.

TABLE III. DETECTOR PERFORMANCE MEASUREMENT

	Pass	Detected
Ransomware	FP	TP
Safe File	TN	FN

Precision focuses on the detector quality which is correctly detected.

$$\text{precision} = \frac{TP}{TP+FP} \quad (1)$$

The recall will calculate correctly detect divided by total file being scan. Another word of recall is true positive rate.

$$\text{Recall} = \frac{TP}{TP+FN} \quad (2)$$

F1 score provides balance between value of precision and recall.

$$F1 \text{ Score} = \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

The accuracy is calculated and carried out by the sum of successfully detect the ransomware and successfully pass the safe file, when divided by total folder being detected.

$$\text{Accuracy} = \frac{TP+FN}{TP+TN+FP+FN} \quad (4)$$

For the system response time, the detector requires 0.12 seconds to 5.41 seconds to fully scan a document according to the complexity of content of the document. So, if a file contains 10 documents, the system will require a minimum of 1.2 seconds to maximum 54.10 seconds to fully scan the file. Table V shows the overall detector performance Detector Performance.

TABLE V. DETECTOR PERFORMANCE

Metric	Performance
Precision	82.86%
Recall	80.56%
F1 Score	81.71%
Accuracy	81.94%
System Response Time	0.12 ~ 5.41 seconds per document

In a nutshell, these metrics provide an overview of the system's effectiveness and efficiency in identifying ransomware threats and avoiding not correctly indicating safe files. The information is important for evaluating the system's reliability and usability of the application.

D. Confusion Matrix

Table VI presents a comparison between state-of-the-art works and the proposed approach.

TABLE VI. COMPARISON AMONG THE STATE-OF-THE ART WORKS

Name	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	Datasets detail
Detector with XGBoost and Random Forest Model [21]	99.61, 99.7	99.74, 99.76	99.38, 99.56	99.56, 99.66	Kaggle dataset (62,485 unique values, 18 functions)
Ransomwar e-Transaction -detection [22]	91.18	89.6	93.19	91.36	UCI dataset (2,916,697 instances)
ClamAV [23]	59.94	50	67	50	416,561 samples
Propsoed Work	81.94	82.86	80.56	81.71	144 files (equal split: ransomwa re vs. safe files)

The proposed work has an accuracy of 81.94%, which is lower than [21] and [22] but significantly better than

ClamAV (59.94%), indicating an improvement over traditional antivirus methods. The proposed model has a recall of 80.56%, ensuring a balanced detection rate but lower than [21] and [22], which leverage machine learning on large datasets. ClamAV's recall (67%) suggests its inability to detect a significant portion of ransomware samples. The dataset size plays a crucial role in model performance. The XGBoost and Random Forest model benefits from a significantly larger dataset, contributing to its higher accuracy. The proposed work uses a much smaller dataset (144 files), which may limit its generalizability and effectiveness against diverse ransomware families. The proposed work demonstrates competitive performance compared to traditional antivirus solutions (ClamAV) but lags behind advanced ML-based models that utilize large datasets. Furthermore, the table illustrates how the proposed work aligns with or diverges from existing methods, showcasing its potential to fill gaps in the current state-of-the-art while maintaining practicality and usability.

V. CONCLUSION

In the project, some effective mitigation plans are suggested from prevention to post-event of the ransomware attack. Strong security measures, more strengthening of machine learning techniques, usually backup of the data, and the implementation of post-event evaluation a greatly effective steps to mitigate the ransomware attack to reduce the damage or prevent from suffering again the attack. Therefore, the main goal is to make people and organizations stronger against ransomware and reduce the damage it manages to cause.

After that, the report showed that the ransomware attack and the ransom amount are growing along with quantity year by year. The target is not only in the retail, finance, and healthcare sectors but also in the other sectors, for example, military, government, and education. Thus, it is essential to protect against ransomware attacks because the target will spread across broader religions of industries, geographies, and networks, and finally become a global threat.

Next, an important part of the project is to create a ransomware detector. The tool is tested based on its accuracy, and the detector response time. The test showed that the detector can detect the ransomware, and how long it takes to finish scanning all the folders which the user inserted.

In a nutshell, the project provides a comprehensive view of how to understand and fight against ransomware. The methods and tools provided can help personnel or organizations reduce the probability of the attack and the damage caused by the ransomware attack. As ransomware keeps changing its pattern, we need to keep improving our defense techniques and plans. Last, the project was created to make the Internet safer, and more peaceful.

The future work includes upgrading the technique of ransomware detection, the technique of compressed file scanning, and the flow of the system. The detector needs to be built for advanced techniques on file searching which belongs to the ransomware detection. The technique that needs to be improved is on the file searching to nested zip files because the technique used to detect the normal file directory and zip file is different, so the detector is not able to scan the nested zip file. After that, the technique of compressed file scanning

will be improved. There are several types of common compressed file formats like a zip file, rar compression file, 7z compression file, password protection 7z compression file, tar compression file, and others. Each type of compression file requires a different technique to read the file by the detector so there is require improving the ability of the detector.

Finally, the flow of the system will also affect the overall detector response time. The response time is based on the flow of the code being executed. For example, if the system uses an inefficient algorithm which means the poor or unsuitable searching method, the response time will be longer than usual. Therefore, the flow of the system also needs to be improved from time to time to adapt to the requirements.

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